

AWS Cloud Quest - Game Description & Instructions

Game Overview

AWS Cloud Quest is an interactive, web-based study game designed to help students prepare for the **AWS Certified Cloud Practitioner (CLF-C02) exam**. Built using **Python and Streamlit**, this game transforms traditional exam preparation into an engaging, gamified learning experience.

The game features **62 comprehensive questions** across four critical AWS exam domains:

- **Cloud Concepts**
-  **Security & Compliance**
-  **Technology**
-  **Billing & Pricing**

How to Play

Getting Started:

1. **Access the Game:** The game is hosted online via Streamlit Cloud. Simply open the provided URL in any web browser.
2. **Welcome Screen:** You'll see the game banner, premise, and a preview of the infrastructure diagram.
3. **Click "START GAME"** to begin your quest.

Gameplay Mechanics:

1. Game Board Layout:

- The board displays four category columns (Cloud Concepts, Security, Technology, Billing)
- Each column has **8 point values** ranging from **100 to 1000 points** (100, 200, 300, 400, 500, 600, 700, 800, 1000)
- Lower points = easier questions; higher points = more challenging questions

2. Selecting Questions:

- Click on any available point value button (★) under your chosen category
- The question automatically appears **at the top of the screen** for easy reading
- The game board remains visible below for reference

3. Answering Questions:

- Read the question carefully
- Select your answer from the multiple-choice options (radio buttons)
- Click "  **Submit Answer**"
- You'll receive **instant feedback**:
 -  **Correct:** Green success message showing points earned and current streak
 -  **Incorrect:** Red error message revealing the correct answer

- An **explanation** appears for every question to reinforce learning

4. Scoring System:

- **Points:** Earn the point value of each correctly answered question (100-1000 pts)
- **Streak Counter:** Build consecutive correct answers to track your momentum
- **Best Streak:** Your highest streak is saved and displayed in the sidebar
- **Progress Tracker:** The sidebar shows total questions answered and current score

5. Completing Questions:

- Once answered, the point value button becomes grayed out and crossed out (~~1000~~)
- This prevents duplicate attempts and helps you track progress
- Click " **Back to Board**" to return to the game board and select another question

6. Winning the Game:

- Complete all **62 questions** across all categories
- Achieve the highest possible score
- When finished, celebratory balloons appear with your final score and statistics

7. Reset Option:

- Click " **Reset Game**" in the sidebar at any time to start over with a clean slate

Learning Objectives

This game helps you master:

- **AWS Global Infrastructure** (Regions, Availability Zones, Edge Locations)
- **Cloud Computing Fundamentals** (Benefits, deployment models, high availability)
- **Security Best Practices** (Shared Responsibility Model, IAM, MFA, compliance)
- **Core AWS Services** (EC2, S3, RDS, Lambda, VPC, CloudFront, etc.)
- **Billing & Cost Management** (Pricing models, Free Tier, Cost Explorer, Budgets)



Visual Elements

The game incorporates **three professional images** to enhance the learning experience:

1. Game Banner (Top of screen):

- Cloud computing visualization with network connections
- Establishes the AWS/cloud theme

2. Infrastructure Diagram (Welcome screen):

- Visual representation of AWS global infrastructure
- Shows Regions, Availability Zones, and Edge Locations

3. **Game Board Preview** (Welcome screen):

- Sample layout showing the category structure
- Helps players understand the game format before starting



How to Start Playing

Option 1: Local Testing (Already Done)

```
streamlit run app.py
```

This opens the game at <http://localhost:8501>

Option 2: Online Access (After Deployment)

- Once deployed to Streamlit Cloud, share the URL with classmates
- Example: <https://github-username-aws-cloud-quest.streamlit.app>
- No installation required - plays in any modern web browser

Technical Specifications

- **Platform:** Python 3.x with Streamlit framework
 - **Hosting:** Streamlit Cloud (free tier)
 - **Browser Compatibility:** Chrome, Firefox, Safari, Edge
 - **Responsive Design:** Adapts to different screen sizes
 - **No Login Required:** Instant access for all students
 - **Mobile-Friendly:** Playable on tablets and smartphones
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